

Exercises

*1. Analyze the logical forms of the following statements:

- (a) We'll have either a reading assignment or homework problems, but we won't have both homework problems and a test. $(R \vee H) \wedge \neg(H \wedge T)$
- (b) You won't go skiing, or you will and there won't be any snow. $\neg S \vee (S \wedge \neg G)$
- (c) $\bar{7} \leq 2$. $\neg(7 < 2) \vee (7 = 2)$

2. Analyze the logical forms of the following statements:

- (a) Either John and Bill are both telling the truth, or neither of them is. $(J \wedge B) \vee \neg(J \wedge B)$
- (b) I'll have either fish or chicken, but I won't have both fish and mashed potatoes. $(F \vee C) \wedge \neg(F \wedge P)$
- (c) 3 is a common divisor of 6, 9, and 15. $6 \wedge 9 \wedge 15$

3. Analyze the logical forms of the following statements:

- (a) Alice and Bob are not both in the room. $\neg(A \wedge B)$
- (b) Alice and Bob are both not in the room. $\neg A \wedge \neg B$
- (c) Either Alice or Bob is not in the room. $\neg A \vee \neg B$
- (d) Neither Alice nor Bob is in the room. $\neg A \wedge \neg B$

4. Analyze the logical forms of the following statements:

- (a) Either both Ralph and Ed are tall, or both of them are handsome. $(R \wedge E) \vee (H \wedge I)$
- (b) Both Ralph and Ed are either tall or handsome. $(R \wedge E) \vee (H \wedge I)$
- (c) Both Ralph and Ed are neither tall nor handsome. $\neg(R \wedge E) \wedge \neg(H \wedge I) = (\neg R \wedge \neg E) \wedge (\neg H \wedge \neg I)$
- (d) Neither Ralph nor Ed is both tall and handsome.

$\bar{7}$ is not less than or equal to 2

4

$\neg(A \wedge B)$
 $= \neg A \vee \neg B$
 (De Morgan's Law)
 $\neg(A \vee B)$
 $= \neg A \wedge \neg B$

Both men are
 & their trucks
 are also
 same

As all is in room

Bob

it is not the case that both are in the room

R is not tall & R is not handsome

5. Which of the following expressions are well-formed formulas?

(a) $\neg(\neg P \vee \neg R)$. $\checkmark$ (correct)

(b) $\neg(P, Q, \wedge R)$. $\times$

(c) $P \wedge \neg P$. $\checkmark$

(d) $(P \wedge Q)(P \vee R)$. $\times$

It is not the case that I will buy pants & I will not buy shirt.

- *6. Let P stand for the statement "I will buy the pants" and S for the statement "I will buy the shirt." What English sentences are represented by the following formulas?

(a) $\neg(P \wedge \neg S)$. I will not buy the pants without the shirt.

(b) $\neg P \wedge \neg S$. I will not buy pants & I will not buy shirt.

(c) $\neg P \vee \neg S$. I will not buy pants OR I will not buy shirt = I will not buy both pants & shirt.

7. Let S stand for the statement "Steve is happy" and G for "George is happy." What English sentences are represented by the following formulas?

(a) $(S \vee G) \wedge (\neg S \vee \neg G)$.

(b) $[S \vee (G \wedge \neg S)] \vee \neg G$.

(c) $S \vee [G \wedge (\neg S \vee \neg G)]$.

8. Let T stand for the statement "Taxes will go up" and D for "The deficit will go up." What English sentences are represented by the following formulas?

(a) $T \vee D$. Taxes will go up or deficit will go up.

(b) $\neg(T \wedge D) \wedge \neg(\neg T \wedge \neg D)$.

(c) $(T \wedge \neg D) \vee (D \wedge \neg T)$.

9. Identify the premises and conclusions of the following deductive arguments and analyze their logical forms. Do you think the reasoning is valid? (Although you will have only your intuition to guide you in answering this last question, in the next section we will develop some techniques for determining the validity of arguments.)

(a) Jane and Pete won't both win the math prize. Pete will win either the math prize or the chemistry prize. Jane will win the math prize. Therefore, Pete will win the chemistry prize.

(b) The main course will be either beef or fish. The vegetable will be either peas or corn. We will not have both fish as a main course and corn as a vegetable. Therefore, we will not have both beef as a main course and peas as a vegetable.

(c) Either John or Bill is telling the truth. Either Sam or Bill is lying. Therefore, either John is telling the truth or Sam is lying.

(d) Either sales will go up and the boss will be happy, or expenses will go up and the boss won't be happy. Therefore, sales and expenses will not both go up.

A common trap in logic:

I will not buy pants & shirt

I will not buy pants & I will not buy shirt.

Both these sentences are different.

$\neg(P \wedge S)$

$\neg P \wedge \neg S$ not same

The first one is commonly interpreted like this.

$\neg(P \wedge S)$

$\neg(a \wedge b)$

I will not do both a and b (Means I can do one or I can do both but not both)

I cannot do both of them

$\neg(a \vee b)$

I will not do a or b (same meaning)

$\neg(a \vee b)$ = It is not the case that I will do a or b.

$B \vee F$

$P \vee C$

$\neg(F \wedge C)$

$\neg(B \wedge P)$ not valid (Doesn't make sense)

English:

I will do neither a nor b

OR

I won't do a & I won't do b

valid: boss cannot be happy & not happy at same time & first statement means that either sales or expenses will go up.

1.2

1) $\sim P \vee Q$

F	T	T	T
F	T	F	F
T	F	T	T
T	F	F	F

2) $\sim [P \wedge (Q \vee \sim P)]$

F	T	T	T	F
T	T	F	F	F
T	F	F	T	T
T	F	F	F	T

3) $(P \vee Q) \wedge (\sim P \vee R)$

T	T	T	T	F	T	T
T	T	F	F	F	F	F
F	F	F	F	F	T	T
F	F	F	F	F	F	F
F	T	T	T	T	T	T
F	T	T	T	T	T	F
F	F	F	F	T	T	T
F	F	F	F	T	T	F

4) (a) $P \leftrightarrow Q$

T	F	T
T	T	F
F	T	T
F	F	F

(b) Because it is true when either P is T and Q is F or Q is T and P is F, so ans is

$$(P \wedge \sim Q) \vee (\sim P \wedge Q)$$

T	F	F	F	F	T
T	T	T	F	F	F
F	F	F	T	T	T
F	F	T	F	T	F

Verified.

4) $\sim (\sim P \wedge \sim Q)$

It is True only when P & Q both are false & we took its negation.

$$\sim (\sim P \wedge \sim Q)$$

T	F	T	F	T
T	F	T	F	F
T	T	F	F	T
F	T	F	T	F

Verified.

5) $P \downarrow Q$

T	F	T
T	F	F
F	F	T
F	T	F

Neither P nor Q means P must be false and Q must be false (same as $\sim P \wedge \sim Q$)

(b) $\sim P \wedge \sim Q$ (as above verified)

(c) $\sim P = P \downarrow P$

P	$\downarrow$	P
T	F	T
T	F	T
F	T	F
F	T	F

also $\sim (P \vee Q) = P \downarrow Q$

either P or Q neither P nor Q

$$P \vee Q = (P \downarrow Q) \downarrow (P \downarrow Q)$$

As $P \downarrow Q$ is negation of $P \vee Q$, so we only needed to negate it once more. To do so we nored it with its elf.

(P	$\downarrow$	Q)	$\downarrow$	(P	$\downarrow$	Q)
T	F	T	T	F	F	T
T	F	F	T	F	F	F
T	F	T	T	F	F	F
F	T	F	F	T	T	F

$$p \wedge q = \neg \neg p \wedge q$$

$$\text{as } \neg(\neg(p \wedge q)) = \neg(\neg p \vee \neg q) \quad (\because \text{double negation law})$$

This is equal

This is equal to $p \wedge q$

for $(p \wedge q)$ as $\neg p \vee \neg q = \neg(p \wedge q)$
 in $\neg(\neg p \vee \neg q)$, we replace $\vee$ by $\wedge$, then it will negate it as
 $\wedge$ is not of OR. so we don't need the external negation
 so $\neg(\neg p \vee \neg q) = \neg p \wedge \neg q = p \wedge q$

$\neg p$	$\downarrow$	$\neg q$
F	T	F
T	F	T
T	F	F
T	F	F

$\neg p \wedge \neg q$ verified

This is correct but we ~~don't~~ can't use $\neg$, so to $\neg p$, we will use $p \downarrow p$
 so formula is $(p \downarrow p) \downarrow (p \downarrow q)$

$$\downarrow \quad \downarrow$$

$$\neg p \downarrow \neg q$$

(a)

P	I	Q	P and Q are not both true
T	F	T	
T	T	F	
F	T	T	
F	T	F	

(b) $p \downarrow q = \neg(p \wedge q)$
 it is false only when both p & q are T, so it must be negation of $p \wedge q$.

(c) $\neg p = p \downarrow p$
 (can be seen from T.T)
 $p \vee q = \neg \neg(p \vee q) = \neg(p \wedge \neg q)$
 as $p \vee q = \neg \neg(p \vee q) = \neg(p \wedge \neg q)$
 if we replace $\neg$ by 1, the external $\neg$ gets cancelled out leaving $\neg p \downarrow \neg q$
 $= (p \downarrow p) \downarrow (q \downarrow q)$

$p \wedge q$
 as $p \downarrow q = \neg(p \wedge q)$, we only need to negate again $\neg(p \downarrow q) = p \wedge q$

or we know that
 $p \wedge q = \neg \neg(p \wedge q)$
 $= \neg(p \downarrow q) = (p \downarrow q) \downarrow (p \downarrow q)$
 $\downarrow$
 by changing $\neg$ to 1, we cancelled one of external $\neg$.

1.2

7) (a) $\neg(J \wedge P)$
 $P \vee C$
 $J \vee C$

J	P	C	$\neg(J \wedge P)$	$P \vee C$	$J \vee C$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	F

Valid

(b) $B \vee F$
 $P \vee C$
 $\neg(F \wedge C)$
 $\therefore \neg(B \wedge P)$

B	F	P	C	$B \vee F$	$P \vee C$	$\neg(F \wedge C)$	$\neg(B \wedge P)$
T	T	T	T	T	T	F	T
T	T	T	F	T	T	T	T
T	T	F	T	T	T	F	T
T	T	F	F	T	F	T	F
T	F	T	T	T	T	T	F
T	F	T	F	T	T	T	F
T	F	F	T	T	T	T	F
T	F	F	F	T	F	T	F
F	T	T	T	T	T	T	F
F	T	T	F	T	T	F	T
F	T	F	T	T	F	T	F
F	T	F	F	T	F	T	F
F	F	T	T	F	T	T	F
F	F	T	F	F	T	T	F
F	F	F	T	F	T	T	F
F	F	F	F	F	F	T	F

Not valid

(c) $J \vee B$
 $\neg S \vee M \vee B$
 $\therefore J \vee \neg S$

J	B	S	$J \vee B$	$\neg S$	$J \vee \neg S$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	T	F	T
T	F	F	T	T	T
F	T	T	T	F	T
F	T	F	T	T	T
F	F	T	F	F	F
F	F	F	F	T	T

Valid

$$(d) (s \wedge h) \vee (e \wedge \neg h)$$

$s \wedge e$

s	h	e	(s ∧ h) ∨ (e ∧ ¬h)
T	T	T	T
T	T	F	T
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

$s \wedge e$

Not valid

$$8) (p \wedge q) \vee (\neg p \wedge \neg q)$$

(a)	p	q	(p ∧ q) ∨ (¬p ∧ ¬q)
	T	T	T
	T	F	F
	F	T	F
	F	F	T

(b)

T

T

I

$$(p \vee \neg q) \wedge (q \vee \neg p)$$

(a)	p	q	(p ∨ ¬q) ∧ (q ∨ ¬p)
	T	T	T
	T	F	F
	F	T	F
	F	F	T

(c)

(d)

(a) = (c), (b) = (d), d is equivalent to none

$$9) (p \vee q) \wedge (\neg p \vee \neg q)$$

(a)	p	q	(p ∨ q) ∧ (¬p ∨ ¬q)
	T	T	F
	T	F	F
	F	T	F
	F	F	T

Neither tautology, nor contradiction

$$(d) [p \wedge (q \vee \neg r)] \vee (\neg p \vee \neg q)$$

	p	q	r	[p ∧ (q ∨ ¬r)] ∨ (¬p ∨ ¬q)
	T	T	T	T
	T	T	F	T
	T	F	T	T
	T	F	F	T
	F	T	T	T
	F	T	F	T
	F	F	T	T
	F	F	F	T

Tautology

10) $\neg(p \vee q) : \neg p \wedge \neg q$

(a)

F	T	T	F	F	T	F	F
F	T	T	F	F	F	F	T
F	T	T	T	T	F	F	F
F	F	T	T	T	F	F	F
T	F	F	F	T	T	T	T

Verified.

1.2

(b)

$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r)$
T	T	T
T	T	F
T	T	T
T	F	F
T	F	F
F	F	F
F	F	F
F	F	F
F	F	F
F	F	F
F	F	F

Verified.

11) (a) $\neg(\neg p \wedge \neg q)$
 $= \neg \neg p \vee \neg \neg q$ (De Morgan)
 $= p \vee q$

(b) $(p \wedge q) \vee (p \wedge \neg q)$
 $= p \wedge (q \vee \neg q)$ $\therefore$ Distributive law
 $= p \wedge T$
 $= p$

(c) $\neg(p \wedge \neg q) \vee (\neg p \wedge q)$

$\neg(p \wedge \neg q) \vee (\neg p \wedge q)$ $\therefore$ De Morgan
 $= (\neg p \vee q) \vee (\neg p \wedge q)$
 We can now apply distributive law treating $\neg p \vee q$ as A $\rightarrow (A \vee \neg p) \wedge (A \vee q)$
 $= ((\neg p \vee q) \vee \neg p) \wedge ((\neg p \vee q) \vee q)$
 $= (\neg p \vee (q \vee \neg p)) \wedge (\neg p \vee (q \wedge q))$
 $\therefore$ Associative laws

$= (\neg p \vee (\neg p \vee q)) \wedge (\neg p \vee q)$
 $\therefore$ Commutative $\therefore$ idempotent law
 $= ((\neg p \vee \neg p) \vee q) \wedge (\neg p \vee q)$
 $\therefore$ Associative law
 $= (\neg(p \wedge p) \vee q) \wedge (\neg p \vee q)$
 $\therefore$ De Morgan
 $= (\neg p \vee q) \wedge (\neg p \vee q)$
 $(\therefore$ idempotent
 $= \neg p \vee q$
 $\therefore$ idempotent

$$\begin{aligned}
 (2) (a) & \neg(\neg p \vee q) \vee (p \wedge \neg r) \\
 & = (p \wedge \neg q) \vee (p \wedge \neg r) \quad \therefore \text{De Morgan} \\
 & = p \wedge (\neg q \vee \neg r) \quad \therefore \text{Distributive} \\
 \text{Note: we can see that } p & \text{ can be taken common so don't try to expand.}
 \end{aligned}$$

$$\begin{aligned}
 (b) & \neg(\neg p \wedge q) \vee (p \wedge \neg r) \\
 & = (p \vee \neg q) \vee (p \wedge \neg r) \quad \therefore \text{De Morgan} \\
 & \text{we cannot take } p \text{ common as it is not a distribution} \\
 & = ((p \vee \neg q) \vee p) \wedge ((p \vee \neg q) \vee \neg r) \quad \therefore \text{Dist} \\
 & = (p \vee (\neg q \vee p)) \wedge ((p \vee \neg q) \vee \neg r) \quad \therefore \text{Ass} \\
 & = (p \vee (p \vee \neg q)) \wedge ((p \vee \neg q) \vee \neg r) \quad \therefore \text{Comm} \\
 & = ((p \vee p) \vee \neg q) \wedge ((p \vee \neg q) \vee \neg r) \quad \therefore \text{Ass} \\
 & = (p \vee \neg q) \wedge ((p \vee \neg q) \vee \neg r) \quad \therefore \text{idempot} \\
 & = p \vee \neg q \quad \therefore \text{Absorp}
 \end{aligned}$$

$$\begin{aligned}
 (c) & (p \wedge r) \vee [\neg r \wedge (p \vee q)] \\
 & = [(p \wedge r) \vee \neg r] \wedge [(p \wedge r) \vee (p \vee q)] \quad \therefore \text{Dist} \rightarrow \text{this is } A \vee (B \vee C) \\
 & \quad \therefore \text{Dis} \quad \text{so we cannot apply dist} \\
 & = [\neg r \vee (p \wedge r)] \wedge [(p \vee q) \vee (p \wedge r)] \quad \therefore \text{Comm} \quad \text{so we first apply comm} \\
 & = [(\neg r \vee p) \wedge (\neg r \vee r)] \wedge [(p \vee q) \vee p] \wedge [(p \vee q) \vee r] \quad \therefore \text{Dist} \\
 & = [(\neg r \vee p) \wedge T] \wedge [(p \vee (q \vee p)) \wedge (p \vee q \vee r)] \quad \therefore \text{Tautology} \\
 & = (\neg r \vee p) \wedge [(p \vee (p \vee q)) \wedge (p \vee q \vee r)] \quad \therefore \text{Associative} \\
 & = (\neg r \vee p) \wedge [(p \vee p) \vee q] \wedge (p \vee q \vee r) \quad \therefore \text{Tautology} \\
 & \quad \text{law} \quad \therefore \text{Comm} \quad \text{because of ass law } (p \vee q) \vee r \text{ is same as } p \vee q \vee r
 \end{aligned}$$

$$\begin{aligned}
 & (\neg r \vee p) \wedge ((p \vee q) \wedge (p \vee q \vee r)) \\
 & \quad \therefore \text{idempotent} \\
 & = (\neg r \vee p) \wedge (p \vee q) \\
 & \quad \therefore \text{Absorp} \\
 & = (p \vee \neg r) \wedge (p \vee q) \\
 & \quad \therefore \text{Comm} \\
 & = p \vee (\neg r \wedge q) \quad \therefore \text{Dist} \\
 & \text{or } p \vee (q \wedge \neg r)
 \end{aligned}$$

Remember if we have $A \wedge (B \wedge C)$, we can apply associative, not distributive.
if we have $A \wedge (B \vee C)$ or $A \vee (B \wedge C)$ then we can apply distributive

$$13) \sim(p \wedge q) = \sim p \vee \sim q$$

Replacing p & q by $\sim p$ & $\sim q$ respectively

$$\sim(\sim p \wedge \sim q) = \sim \sim p \vee \sim \sim q$$

$$\sim(\sim p \wedge \sim q) = p \vee q$$

Applying negation on both sides

$$\sim \sim(\sim p \wedge \sim q) = \sim(p \vee q)$$

$$\sim p \wedge \sim q = \sim(p \vee q)$$

$$\text{or } \sim(p \vee q) = \sim p \wedge \sim q$$

If one of premise is tautology, we cannot say about validity as it depends on other premises and conclusion.

$$14) [p \wedge (q \wedge r)] \wedge s$$

$$= [(p \wedge q) \wedge r] \wedge s \therefore \text{Ass}$$

$$= (p \wedge q) \wedge [r \wedge s] \therefore \text{Ass}$$

$$= (p \wedge q) \wedge (r \wedge s)$$

$$= ((p \wedge q) \wedge r) \wedge s$$

$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ (A \wedge B) \wedge C \\ = A \wedge (B \wedge C) \end{matrix}$$

If a premise is a contradiction there will be no case where all premises are true at same time & argument is vacuously valid.

Note: By definition an argument is valid if the premises cannot be all true without the conclusion being true as well

15) A truth table having n letters (statements) will have 2^n rows

18) If ~~all~~ conclusion is a tautology, it will be true whenever premises are all true, so it is valid.

If conclusion is a contradiction, it will be invalid unless there is no case where all premises are true at same time. Then it is vacuously valid.

1) (a) let $P(y, x) = y$ is divisor of x
 then this statement is $P(3, 6) \wedge P(3, 9) \wedge P(3, 15)$

(b) $P(x, y) = x$ is divisible by y
 $[P(x, 2) \wedge P(x, 3)] \wedge \neg P(x, 4)$ or $P(x) = 3$ divides y
 $P(6) \wedge P(9) \wedge \neg P(15)$

(c) $N(x) = x$ is natural number, $P(y) = y$ is ~~prime~~ ^{prime} number
 $(P(x) +^x P(y))$ ($\therefore$ exclusive OR, x or y but not both)
 $N(x) \wedge N(y) \wedge (P(x) + P(y))$

or $(P(x) \wedge \neg P(y)) \vee (\neg P(x) \wedge P(y))$

2) (a) $M(x) = x$ is man $P(x, y) = x$ is taller than y
 $M(x) \wedge M(y) \wedge [P(x, y) \vee P(y, x)]$

(b) $B(x) = x$ has brown eyes, $R(x) = x$ has red hair
 $[B(x) \vee B(y)] \wedge [R(x) \vee R(y)]$

(c) $[B(x) \wedge R(x)] \vee [B(y) \wedge R(y)]$

3) (a) $x \in \{p \mid p \text{ is a planet in solar system}\}$

(b) $\{x \mid x \text{ is an Ivy league school}\}$

(c) $\{x \mid x \text{ is a state in United States}\}$

(d) $\{x \mid x \text{ is a province or territory in Canada}\}$

4) (a) $\{x \in \mathbb{N} \wedge (x \neq 1) \mid x \neq 0 \mid x \times x\}$

(b) $\{x \in \mathbb{N}^+ \mid 2^x\}$

= both of these were infinite sets

(c) $\{x \mid x \geq 10 \wedge x \leq 19\}$

5) $\{x \in \mathbb{R} \mid 13 - 2x > 1\}$ means x is real number such that $13 - 2x > 1$. Substituting this in $-3 \in x$ means -3 is a real number such that $13 - 2(-3) > 1$.

No variables are free as $13 + 6 > 1$ so it is True.

Its simplified form is $(-3 \in \mathbb{R}) \wedge (13 - 2(-3) > 1)$

(b) $(4 \in \mathbb{R}) \wedge (13 - 2(4) > 1)$, No free variables & this statement is False

(c) $(5 \notin \mathbb{R}) \wedge (13 - 2(5) > c)$, c is a free variable
 $\wedge [(5 \in \mathbb{R}) \wedge (13 - 2(5) > c)]$ x is bound

6) $(w \in \mathbb{R}) \wedge (13 - 2w > c)$ w & c are free, x is bound

(b) $(4 \in \mathbb{R}) \wedge (13 - 2(4) \in P)$ No free variables (True)

(c) $(4 \in P) \wedge (13 - 2(4) > 1)$ No free variables (False)

7) (a) ~~$\{x \in \mathbb{R}\}$~~

7. List the elements of the following sets:

(a) $\{x \in \mathbb{R} \mid 2x^2 + x - 1 = 0\}$. $\{1\}$

(b) $\{x \in \mathbb{R}^+ \mid 2x^2 + x - 1 = 0\}$. $\emptyset$

(c) $\{x \in \mathbb{Z} \mid 2x^2 + x - 1 = 0\}$. $\{-1\}$

(d) $\{x \in \mathbb{N} \mid 2x^2 + x - 1 = 0\}$. $\emptyset$

*8. What are the truth sets of the following statements? List a few elements of the truth set if you can.

(a) Elizabeth Taylor was once married to x .

(b) x is a logical connective studied in Section 1.1. $\{\wedge, \vee, \neg\}$

(c) x is the author of this book. $\{\text{Daniel J. Velleman}\}$

9. What are the truth sets of the following statements? List a few elements of the truth set if you can.

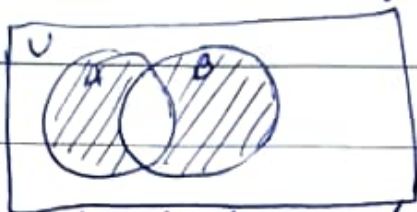
(a) x is a real number and $x^2 - 4x + 3 = 0$. $\{1, 3\}$

(b) x is a real number and $x^2 - 2x + 3 = 0$. $\emptyset$

(c) x is a real number and $5 \in \{y \in \mathbb{R} \mid x^2 + y^2 < 50\}$. $\{\}$ or $\emptyset$

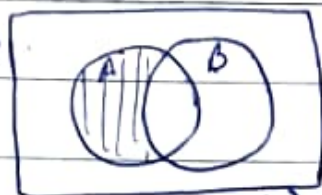
$25 + 25 \neq 50$

3)



$$(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$$

4)(a)



$$A \setminus (A \cap B) = A \setminus B$$

(b)



$$A \setminus (B \cap C)$$

Proved!



$$(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$$

$$\begin{aligned} 5) A \setminus (A \cap B) &= x \in A \wedge \neg(x \in A \wedge x \in B) && \therefore \text{De Morgan} \\ &= x \in A \wedge (x \notin A \vee x \notin B) && \therefore \\ &= (x \in A \wedge x \notin A) \vee (x \in A \wedge x \notin B) && \therefore \text{Dist} \\ &= F \vee (x \in A \wedge x \notin B) && \therefore \text{Contrad} \\ &= (x \in A \wedge x \notin B) && \therefore \text{Contrad} \\ &= A \setminus B && \therefore \text{By definition} \end{aligned}$$

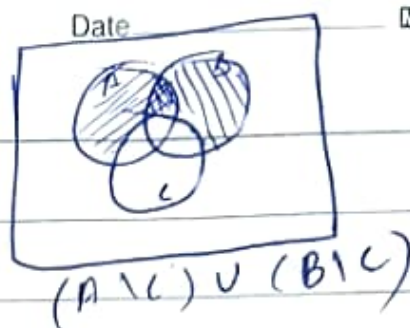
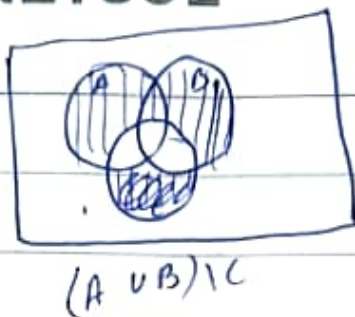
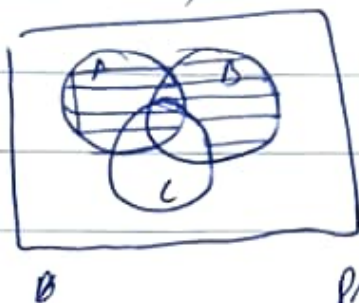
$$(b) A \cup (B \cap C) = x \in A \vee (x \in B \wedge x \in C)$$

$$= (x \in A \vee x \in B) \wedge (x \in A \vee x \in C) \therefore \text{Dist}$$

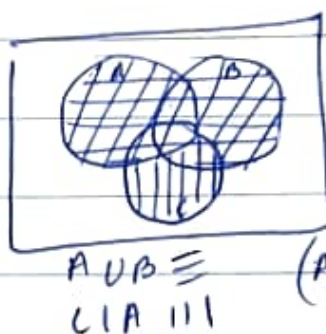
$$= (A \cup B) \cap (A \cup C) \therefore \text{By def}$$

$$= (A \cup B) \cap (A \cup C) \therefore \text{By def}$$

6) (a)


(b) $A \cup (B \setminus C)$


proved!



$$A \cup B \equiv C \setminus A$$

$$(A \cup B) \setminus (C \setminus A) = \text{shaded}$$

$$1) (A \cup B) \setminus C = (x \in A \vee x \in B) \wedge \neg(x \in C)$$

$$= \neg(x \in C) \wedge (x \in A \vee x \in B) \quad \because \text{Comm}$$

$$= (\neg(x \in C) \wedge x \in A) \vee (\neg(x \in C) \wedge x \in B) \quad \because \text{Dist}$$

$$= (x \in A \wedge \neg(x \in C)) \vee (x \in B \wedge \neg(x \in C)) \quad \because \text{Comm}$$

$$= (A \setminus C) \vee (B \setminus C) \quad \because \text{By def.}$$

$$= (A \setminus B) \cup (B \setminus C) \quad \because \text{By def.}$$

(b) $A \cup (B \setminus C)$

$$= x \in A \vee (x \in B \wedge \neg(x \in C))$$

$$= (x \in A \vee x \in B) \wedge (x \in A \vee \neg(x \in C))$$

$$= \neg(x \in A \vee \neg(x \in C))$$

$$= (x \in A \vee x \in B) \wedge \neg(x \in A \vee \neg(x \in C)) \quad \because \text{Double negation}$$

$$= (x \in A \vee x \in B) \wedge \neg(x \in A \wedge \neg(x \in C)) \quad \because \text{De Morgan}$$

$$= (A \cup B) \wedge \neg(x \in C \wedge \neg(x \in A)) \quad \because \text{By def.} \quad \because \text{Comm}$$

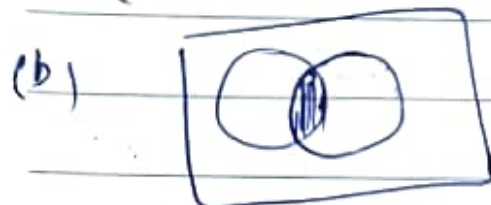
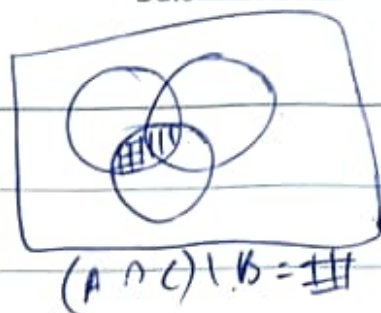
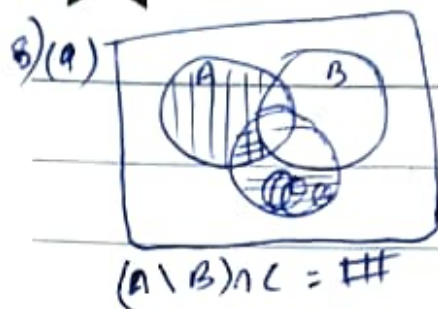
$$= (A \cup B) \wedge \neg(C \setminus A)$$

$$= (A \cup B) \setminus (C \setminus A)$$

By def.

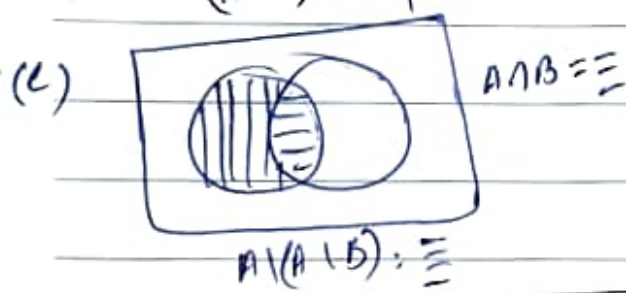
By def.

we conditional $x \in A \vee x \in B$ to $A \cup B$ early for convenience but we should do it at end



$$A \cap B = \text{shaded region}$$

$$(A \cap B) \setminus B = \emptyset$$



$$A \cap B = \text{shaded region}$$

9) (a) $(A \setminus B) \setminus C = (x \in A \wedge x \notin B) \wedge x \notin C$
 $\therefore x \in A \wedge x \notin B \wedge x \notin C$

(b) $A \setminus (B \cap C) = x \in A \wedge \neg(x \in B \cap x \in C)$
 $= x \in A \wedge (x \notin B \vee x \notin C) \therefore \text{De Morgan}$

(c) $(A \setminus B) \cup (A \cap C) = (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in C)$
 $= x \in A \wedge (x \notin B \vee x \in C)$

(d) $(A \setminus B) \cap (A \setminus C) = (x \in A \wedge x \notin B) \cap (x \in A \wedge x \notin C)$
 $= x \in A \wedge x \notin B \wedge x \notin C \therefore \text{Associative}$
 $= x \in A \wedge x \notin B \wedge x \notin C \therefore \text{Idempotent}$

(e) $A \setminus (B \cup C) = x \in A \wedge \neg(x \in B \vee x \in C)$
 $= x \in A \wedge x \notin B \wedge x \notin C \therefore \text{De Morgan}$

(a), (d), (e) are same, b and c are same

10) (a) if $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$

then $(A \cup B) \setminus B = \{1, 2, 3, 4\} - \{2, 3, 4\} = \{1\} \neq A$ but $\subseteq A$

(b) $(A \cup B) \setminus B = (x \in A \vee x \in B) \wedge x \notin B$

$= x \notin B \wedge (x \in A \vee x \in B) \quad \therefore \text{Comm}$

$= (x \notin B \wedge x \in A) \vee (x \notin B \wedge x \in B) \quad \therefore \text{Dist}$

$= (x \in A \wedge x \notin B) \vee F$

$\therefore \text{Comm Contradiction}$

$= A \setminus B$

Proved

By def.

11) $(A \setminus B) \cup B = (x \in A \wedge x \notin B) \vee x \in B$

$= x \in B \vee (x \in A \wedge x \notin B)$

$= (x \in B \vee x \in A) \wedge (x \in B \vee x \notin B)$

$= (x \in A \vee x \in B) \wedge T$

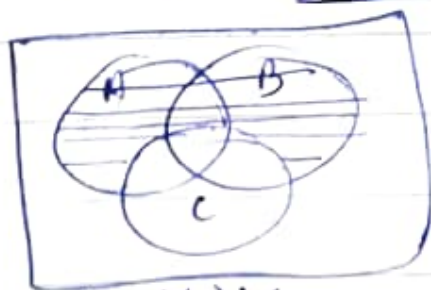
$= x \in A \vee x \in B = A \cup B$

So it's not equal to A

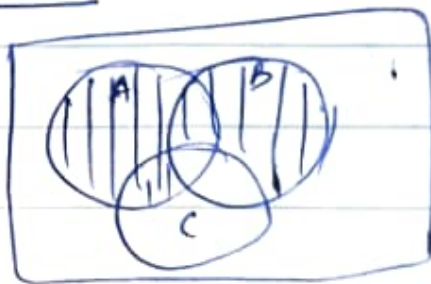
Yes A is subset of $(A \setminus B) \cup C = A \cup B$

It's always equal to $A \cup B$.

13) (a)



$(A \cup B) \setminus C$



$A \cup (B \setminus C)$

This is subset of other

Yes

(b) if $A = \{1, 2, 3\}$, $B = \{2\}$, $C = \{3, 4, 5\}$

then $(A \cup B) \cap C = \{1, 2, 3\} \cap \{3, 4, 5\} = \{3\}$

but $A \cup (B \cap C) = \{1, 2, 3\} \cup \{2\} = \{1, 2, 3\}$

So they are not equal (The key lies in Venn diagram where A & C are overlapping, there is difference).

15)



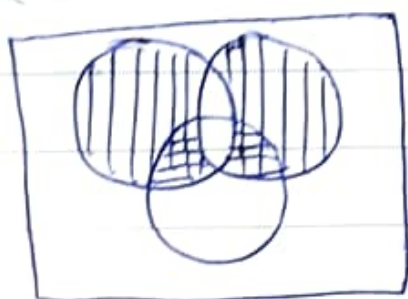
$$(A \cap B) \cup C = \text{||||}$$

same.

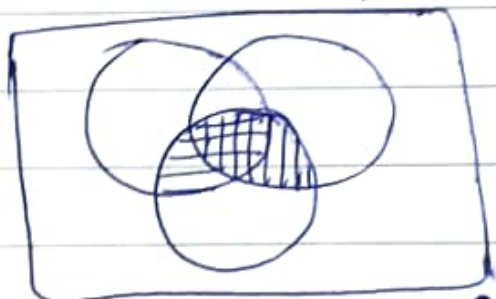


$$(A \cup C) \cap (B \cap C) = \text{||||} = \text{||||}$$

16)

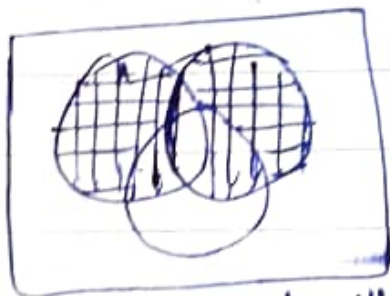


$$(A \cap B) \cap C = \text{##}$$

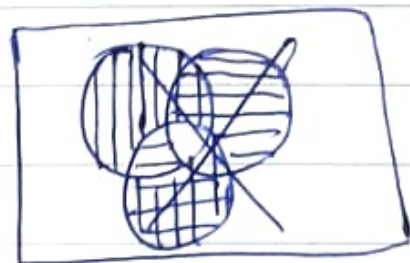


$$(A \cap C) \cap (B \cap C) = \text{##} = \text{##}$$

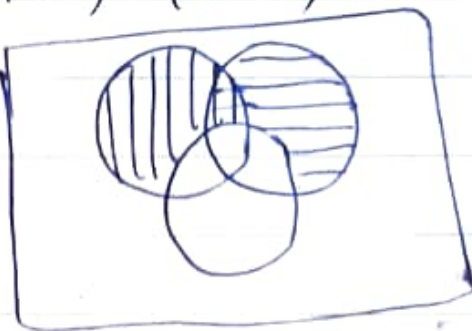
17)



$$(A \cap B) \setminus C = \text{##}$$



$$(A \setminus C) \cap (B \setminus C) = \text{##}$$





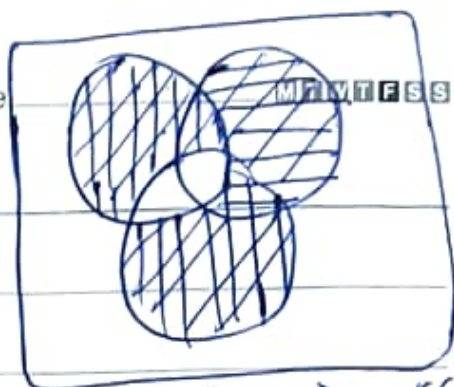
NETSOL

16(a)



$$(A \cup B) \setminus C = \text{grid}, \text{vertical lines}$$

Date



$$(A \setminus C) \cup (B \setminus A) = \text{diagonal lines, cross-hatching}$$

$$A \setminus C = \text{vertical lines}$$

$$B \setminus A = \text{grid}$$

17

*14. Use Venn diagrams to show that the associative law holds for symmetric difference; that is, for any sets A , B , and C , $A \oplus (B \oplus C) = (A \oplus B) \oplus C$.

15. Use any method you wish to verify the following identities:

(a) $(A \oplus B) \cup C = (A \cup C) \oplus (B \setminus C)$.

(b) $(A \oplus B) \cap C = (A \cap C) \oplus (B \cap C)$.

(c) $(A \oplus B) \setminus C = (A \setminus C) \oplus (B \setminus C)$.

16. Use any method you wish to verify the following identities:

(a) $(A \cup B) \oplus C = (A \oplus C) \oplus (B \setminus A)$.

(b) $(A \cap B) \oplus C = (A \oplus C) \oplus (A \setminus B)$.

(c) $(A \setminus B) \oplus C = (A \oplus C) \oplus (A \cap B)$.

17. Fill in the blanks to make true identities:

(a) $(A \oplus B) \cap C = (C \setminus A) \oplus \underline{(C \setminus B)}$

(b) $C \setminus (A \oplus B) = (A \cap C) \oplus \underline{(C \setminus B)}$

(c) $(B \setminus A) \oplus C = (A \oplus C) \oplus \underline{(A \cup B)}$

— This can be done by making Venn diagrams & careful analysis. If we want to exclude a region, we should make it common in both terms & to include it, it should be in one of the 2 terms.

1.5. The Conditional and Biconditional Connectives

(by def of symmetric difference pg 38)

It is time now to return to a question we left unanswered in Section 1.1. We

Exercises

*1. Analyze the logical forms of the following statements:

(a) If this gas either has an unpleasant smell or is not explosive, then it isn't hydrogen. $(U \vee E) \rightarrow \neg H$

(b) Having both a fever and a headache is a sufficient condition for George to go to the doctor. $(F \wedge H) \rightarrow D$

(c) Both having a fever and having a headache are sufficient conditions for George to go to the doctor. $(F \rightarrow D) \wedge (H \rightarrow D)$

(d) If $x \neq 2$, then a necessary condition for x to be prime is that x be odd. $A \rightarrow (P(x) \rightarrow O(x))$

2. Analyze the logical forms of the following statements:

(a) Mary will sell her house only if she can get a good price and find a nice apartment. $S \rightarrow (P \wedge N)$

(b) Having both a good credit history and an adequate down payment is a necessary condition for getting a mortgage. $M \rightarrow (C \wedge D)$

(c) John will drop out of school, unless someone stops him. (Hint: First try to rephrase this using the words *if* and *then* instead of *unless*.)
 If someone doesn't stop John, he will drop out of school. $D \rightarrow S$

(d) If x is divisible by either 4 or 6, then it isn't prime. $(D(x,4) \vee D(x,6)) \rightarrow \neg P(x)$
 This is of form $q \rightarrow \neg p$

3. Analyze the logical form of the following statement:

(a) If it is raining, then it is windy and the sun is not shining.

$$R \rightarrow (W \wedge \neg S)$$

Now analyze the following statements. Also, for each statement determine whether the statement is equivalent to either statement (a) or its converse.

- (b) It is windy and not sunny only if it is raining. $(W \wedge \neg S) \rightarrow R$ (converse of A)
 (c) Rain is a sufficient condition for wind with no sunshine. $R \rightarrow (W \wedge \neg S)$ (equivalent)
 (d) Rain is a necessary condition for wind with no sunshine. $(W \wedge \neg S) \rightarrow R$ (converse)
 (e) It's not raining, if either the sun is shining or it's not windy. $(S \vee \neg W) \rightarrow \neg R$

lack of
sunshine is also
a necessary
condition

- (f) Wind is a necessary condition for it to be rainy, and so is a lack of sunshine. $(R \rightarrow W) \wedge (R \rightarrow \neg S)$
 (g) Either it is windy only if it is raining, or it is not sunny only if it is raining. $(W \rightarrow R) \vee (\neg S \rightarrow R)$

*4. Use truth tables to determine whether or not the following arguments are valid:

- (a) Either sales or expenses will go up. If sales go up, then the boss will be happy. If expenses go up, then the boss will be unhappy. Therefore, sales and expenses will not both go up.
 (b) If the tax rate and the unemployment rate both go up, then there will be a recession. If the GDP goes up, then there will not be a recession. The GDP and taxes are both going up. Therefore, the unemployment rate is not going up.
 (c) The warning light will come on if and only if the pressure is too high and the relief valve is clogged. The relief valve is not clogged. Therefore, the warning light will come on if and only if the pressure is too high.

5. Use truth tables to determine whether or not the following arguments are valid:

- (a) If Jones is convicted then he will go to prison. Jones will be convicted only if Smith testifies against him. Therefore, Jones won't go to prison unless Smith testifies against him. *q unless p*
 (b) Either the Democrats or the Republicans will have a majority in the Senate, but not both. Having a Democratic majority is a necessary condition for the bill to pass. Therefore, if the Republicans have a majority in the Senate then the bill won't pass.

6. (a) Show that $P \leftrightarrow Q$ is equivalent to $(P \wedge Q) \vee (\neg P \wedge \neg Q)$.

(b) Show that $(P \rightarrow Q) \vee (P \rightarrow R)$ is equivalent to $P \rightarrow (Q \vee R)$.

*7. (a) Show that $(P \rightarrow R) \wedge (Q \rightarrow R)$ is equivalent to $(P \vee Q) \rightarrow R$.

(b) Formulate and verify a similar equivalence involving $(P \rightarrow R) \vee (Q \rightarrow R)$.

8. (a) Show that $(P \leftrightarrow Q) \wedge (Q \rightarrow R)$ is equivalent to $(P \rightarrow R) \wedge [(P \leftrightarrow Q) \vee (R \leftrightarrow Q)]$.

(b) Show that $(P \rightarrow Q) \vee (Q \rightarrow R)$ is a tautology.

4) $sve, s \rightarrow h, e \rightarrow nh, \therefore N(sac)$

(a)

T	T	T	T	F	F	F	T
T	T	T	F	T	T	F	T
T	T	F	T	T	F	T	F
T	T	F	F	T	T	T	F
F	T	T	T	F	F	T	F
F	T	T	T	F	F	T	F
F	T	T	F	T	T	T	F
F	F	F	T	T	F	T	F
F	F	F	T	F	T	T	F
F	F	F	F	T	T	T	F

Valid.

b) $(t \wedge u) \rightarrow h, g \rightarrow nh, gnt \therefore nu$

T	T	T	T	T	F	F	F	T	F
T	T	T	T	T	F	T	F	F	F
T	T	T	F	F	T	T	T	T	F
T	T	T	F	F	F	T	T	F	F
T	F	F	T	T	T	F	F	T	T
T	F	F	T	T	F	T	F	F	T
T	F	F	T	F	T	T	T	T	T
T	F	F	T	F	F	T	T	F	T
F	F	T	T	T	T	F	F	F	F
F	F	T	T	F	T	T	T	F	F
F	F	T	T	F	F	T	T	F	F
F	F	T	T	F	T	T	T	F	F
F	F	T	T	F	T	F	F	F	T
F	F	F	T	T	F	T	F	F	T
F	F	F	T	F	T	T	T	F	T
F	F	F	T	F	F	T	T	F	T

Valid.

(c) $W \leftrightarrow (p \wedge q)$, $W \leftrightarrow p$

T	T	T	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
T	F	F	F	T	F
F	F	T	T	F	F
F	T	T	F	T	F
F	T	F	F	F	T
F	T	F	F	T	T

Invalid.

5) $C \rightarrow P$, $C \rightarrow T$, $\therefore NT \rightarrow NP$

T	T	T	T	F	T	✓	F
T	T	T	F	T	F	F	F
T	F	F	T	T	F	T	T
T	F	F	F	F	F	T	T
F	T	T	T	T	T	F	T
F	T	T	T	F	F	T	F
F	T	F	T	T	T	F	T
F	T	F	F	T	F	T	T

Invalid

(b) $D \leftrightarrow R$, $B \rightarrow D$, $R \rightarrow NB$

T	F	T	T	F	F
T	F	T	F	T	T
T	T	F	T	T	✓
T	T	F	F	T	T
F	T	T	T	F	F
F	T	T	F	T	T
F	F	F	T	F	F
F	F	F	F	T	T

Valid

$$\begin{aligned}
 6) \quad p \leftrightarrow q &= (p \rightarrow q) \wedge (q \rightarrow p) = (\neg p \vee q) \wedge (\neg q \vee p) \\
 (a) \quad &= ((\neg p \vee q) \wedge \neg q) \vee ((\neg p \vee q) \wedge p) \quad \text{Dist} \\
 &= (\neg q \wedge (\neg p \vee q)) \vee (p \wedge (\neg p \vee q)) \quad \text{Comm} \\
 &= ((\neg q \wedge \neg p) \vee (\neg q \wedge q)) \vee ((p \wedge \neg p) \vee (p \wedge q)) \quad \text{Dist} \\
 &= (\neg q \wedge \neg p) \vee F \vee (F \vee (p \wedge q)) \quad \text{Contrad} \\
 &= (\neg p \wedge \neg q) \vee (p \wedge q) \quad \text{Comm} \\
 &= (p \wedge q) \vee (\neg p \wedge \neg q) \quad \text{Comm}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad (p \rightarrow q) \vee (p \rightarrow r) \\
 &= (\neg p \vee q) \vee (\neg p \vee r) \\
 &= [\neg p \vee (q \vee r)] \vee r \quad (\because \text{section 1.2 Q14}) \\
 &= [(\neg p \vee \neg p) \vee q] \vee r \quad \text{1. Abs Comm} \\
 &= (\neg p \vee q) \vee r \quad \text{Idempotent} \\
 &= \neg p \vee (q \vee r) \quad \text{Comm} \\
 &= p \rightarrow (q \vee r) \quad \vee p \rightarrow q \vee p \rightarrow r
 \end{aligned}$$

$$\begin{aligned}
 7) (a) \quad (p \rightarrow \neg q) \wedge (q \rightarrow r) \\
 &= (\neg p \vee r) \wedge (\neg q \vee r) \\
 &= (r \vee \neg p) \wedge (r \vee \neg q) \quad \text{Comm} \\
 &= r \vee (\neg p \wedge \neg q) \\
 &= r \vee \neg(p \vee q) \\
 &= \neg(p \vee q) \vee r \\
 &= (p \vee q) \rightarrow r
 \end{aligned}$$

$$(b) (p \rightarrow r) \vee (q \rightarrow r)$$

$$= (\neg p \vee r) \vee (\neg q \vee r)$$

$$= (r \vee \neg p) \vee (r \vee \neg q) \quad \because \text{Commutative}$$

$$= [(r \vee \neg p) \vee r] \vee \neg q \quad (\because \text{sub 1-2 Q14})$$

$$= [(r \vee r) \vee \neg p] \vee \neg q$$

$$= (r \vee \neg p) \vee \neg q \quad \because \text{Idempotent}$$

$$= r \vee (\neg p \vee \neg q) \quad \because \text{Assoc}$$

$$= r \vee \neg(p \wedge q) \quad \because \text{De Morgan}$$

$$= \neg(p \wedge q) \vee r \quad \because \text{Commutative}$$

$$= (p \wedge q) \rightarrow r$$

$$8) (a) (p \rightarrow q) \wedge (q \rightarrow r), (p \rightarrow r) \wedge [(p \leftrightarrow q) \vee (r \leftrightarrow q)]$$

T	T	T	T	T	T	T	T	T	T	T	T	T	T	T
T	T	T	F	F	F	F	F	T	T	F	F	T	T	F
T	F	F	F	T	T	T	T	F	F	F	F	F	F	F
T	F	F	F	T	F	F	F	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T	T	T	F	T	T	T	T
F	T	T	F	F	F	F	F	T	F	F	F	F	F	F
F	T	F	T	T	T	T	T	T	T	T	T	T	F	F
F	T	F	T	T	F	F	F	T	T	T	T	T	T	T
<u> </u>			<u> </u>			<u> </u>			<u> </u>			<u> </u>		

Equivalent

$$(b) (p \rightarrow q) \vee (q \rightarrow r)$$

T	T	T
T	T	F
F	T	T
F	T	T
T	T	T
T	T	F
T	T	T
T	T	T

Yes, it's tautology

$$1) (a) (p \vee q) \leftrightarrow q$$

$$= ((p \vee q) \rightarrow q) \wedge (q \rightarrow (p \vee q))$$

$$= (\neg(p \vee q) \vee q) \wedge (\neg q \vee (p \vee q))$$

$$= ((\neg p \wedge \neg q) \vee q) \wedge ((\neg q \vee q) \vee p) \quad \therefore \text{distributive law}$$

$$= (q \vee (\neg p \wedge \neg q)) \wedge (T \vee p) \quad \therefore \text{Commutative}$$

$$= [(q \vee \neg p) \wedge (q \vee \neg q)] \wedge T \quad \therefore \text{Tautology}$$

$$= (q \vee \neg p) \wedge T$$

$\therefore$ Independent $\therefore$ Tautology

$\therefore$ idempotent

$$= q \vee \neg p$$

$$= \neg p \vee q = p \rightarrow q$$

$$(b) (p \wedge q) \leftrightarrow q$$

$$= ((p \wedge q) \rightarrow q) \wedge (q \rightarrow (p \wedge q))$$

$$= (\neg(p \wedge q) \vee q) \wedge (\neg q \vee (p \wedge q))$$

$$= (\neg p \vee \neg q \vee q) \wedge ((\neg q \vee p) \wedge (\neg q \vee q))$$

$$= (\neg p \vee T) \wedge ((\neg q \vee p) \wedge T)$$

$$= T \wedge (\neg q \vee p)$$

$$= \neg q \vee p = q \rightarrow p$$

The Conditional and Biconditional Connectives

- *9. Find a formula involving only the connectives $\neg$ and $\rightarrow$ that is equivalent to $P \wedge Q$. *$\neg(P \rightarrow \neg Q)$*
10. Find a formula involving only the connectives $\neg$ and $\rightarrow$ that is equivalent to $P \leftrightarrow Q$. *$(\neg Q \rightarrow P) \rightarrow \neg(P \rightarrow \neg Q)$*
11. (a) Show that $(P \vee Q) \leftrightarrow Q$ is equivalent to $P \rightarrow Q$.
 (b) Show that $(P \wedge Q) \leftrightarrow Q$ is equivalent to $Q \rightarrow P$.
12. Which of the following formulas are equivalent?
- (a) $P \rightarrow (Q \rightarrow R)$.
 - (b) $Q \rightarrow (P \rightarrow R)$.
 - (c) $(P \rightarrow Q) \wedge (P \rightarrow R)$.
 - (d) $(P \wedge Q) \rightarrow R$.
 - (e) $P \rightarrow (Q \wedge R)$.

we can easily show my truth table